

## Quiz 1

**Scenario:** You are recruiting a central defender for a team that plays a high defensive line.

**Budget Constraints:**

- Transfer Budget: €45M

**Individual Rule:** Let  $d$  be the last digit of your Student ID.

$$\text{FeeA} = 40 + d$$

$$\text{FeeB} = 35$$

$$\text{FeeC} = 65$$

$$\text{FeeD} = 50 - d$$

A player is financially feasible only if:

$$\text{Fee} < 45$$

**Per 90 Statistics:**

| Player | Tackles Won | Interceptions | Prog. Passes | Penalties Taken |
|--------|-------------|---------------|--------------|-----------------|
| A      | 2.9         | 1.8           | 5.4          | 0.10            |
| B      | 3.6         | 2.4           | 4.2          | 0.00            |
| C      | 2.1         | 1.5           | 6.8          | 0.05            |
| D      | 3.2         | 2.0           | 5.9          | 0.02            |

Assume higher values indicate better performance.

### Questions

- (a) (3 marks) Identify the financially feasible players.
- (b) (4 marks)
- Identify the irrelevant metric
  - Compute percentile ranks
  - Sketch a radar plot
- (c) (3 marks) Recommend one player with justification.

# Solution

## (a) Financially Feasible Players

- Player A:  $\text{€}(40 + d) \rightarrow$  Feasible if  $d < 5$
- Player B:  $\text{€}35 \rightarrow$  Feasible
- Player C:  $\text{€}65 \rightarrow$  Not feasible
- Player D:  $\text{€}(50 - d) \rightarrow$  Feasible if  $d > 5$

### Shortlisted Players:

- Always: B
- If  $d < 5$ : A and B
- If  $d > 5$ : B and D

## (b) Radar Plot Preparation

### Irrelevant metric:

- Penalties Taken

### Relevant metrics:

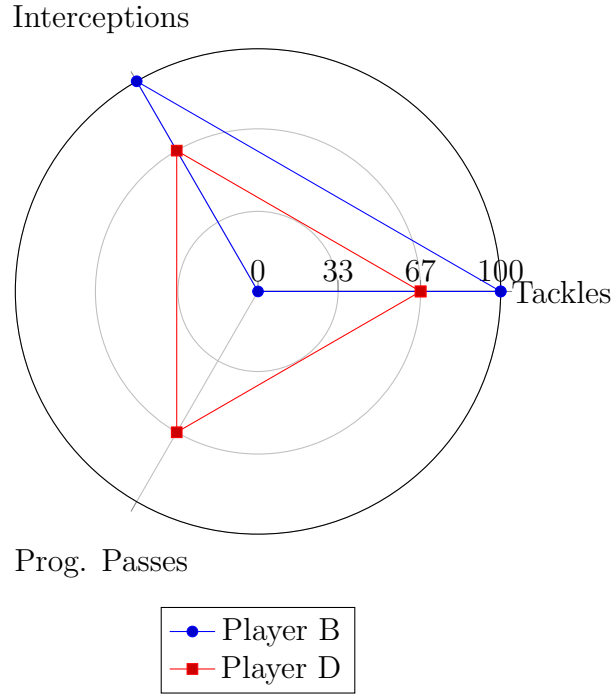
- Tackles Won
- Interceptions
- Progressive Passes

### Percentile formula (Rank 1 = Best):

$$\text{Percentile} = \frac{4 - \text{Rank}}{4 - 1} \times 100$$

### Percentiles (Example: B vs D)

| Player | Tackles | Interceptions | Prog. Passes |
|--------|---------|---------------|--------------|
| B      | 100%    | 100%          | 0%           |
| D      | 67%     | 67%           | 67%          |



### (c) Final Recommendation

#### Recommended Player: B

Player B provides the strongest defensive profile with the highest tackles and interceptions, making him ideal for a high defensive line. Additionally, he is always within budget (€35M), ensuring financial feasibility.

#### Alternative (if $d > 5$ ): Player D

Player D offers a more balanced profile with strong progressive passing while maintaining solid defensive performance. He is a suitable option when ball progression from the back is prioritized.

#### Conclusion:

Choose Player B for defensive solidity and reliability. Choose Player D (if feasible) for a more balanced, ball-playing defender.